Learning objectives

After completing this problem set, students should be able to:

- Approximate the solution of first order initial value problems using the method of dominant balances, at large t or near the location where the solution diverges.
- Generate solutions to first order initial value problems in Matlab/Python using `ode45`.
- Write short functions to perform a certain task in Matlab/Python, for example, implementing Euler's method.
- Transform an initial value problem into a different initial value problem given a specified change of variables relationship.
- Use the existence and uniqueness theorem for second order linear initial value problem to determine the interval in which the problem is guaranteed to have a unique solution.
- Determine when two functions form a linearly independent set, using both the definition of a linearly independent set and the Wronskian determinant.
- Discuss the structure of the general solution to $L[y] = 0$.

Grading

Problems 1, 3 - 8 will be graded for completeness, while problems 2 and 9 will be graded for accuracy and clarity of both mathematics, code, and writing.

Problems

Read sections 2.7, 3.1 - 3.5 of *Elementary Differential Equations and Boundary Value Problems*, by William Boyce and Richard DiPrima. The following problems are based on those sections and material discussed in classes 6-8 and recitation section 3.

1. *Identifying dominant terms*
   For large t, identify the dominant terms in each of the following expressions:

   (a) $7 + t^2$
   (b) $3 + \frac{e^t}{t}$
   (c) $1 + te^{-t^2}$

2. *Approximate solutions*
   For each of the initial value problems below:

   - Determine how the solution $y(t)$ behaves at large t. If the solution $y(t)$ diverges at finite t then describe the functional form of the divergence.
   - Solve the initial value problem numerically in Matlab/Python, using `ode45`. You may use the code provided in class 7 on as a template, posted on the Canvas page.
   - Compare the numerical `ode45` solution at large t to the approximate solution at large t, by plotting your approximate solution on the same figure as the numerical solution.

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a) t^2
b) e^t
c) 1

2. *Approximate solutions*
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- Solve the initial value problem numerically in Matlab/Python, using `ode45`. You may use the code provided in class 7 on as a template, posted on the Canvas page.
- Compare the numerical `ode45` solution at large t to the approximate solution at large t, by plotting your approximate solution on the same figure as the numerical solution.
- As needed, refine your approximate solution until it qualitatively agrees with the numerical solution.

(a)
$$\frac{dy}{dt} + y^2 = \frac{3}{t^3 + \cos t}, \quad y(1) = 2.$$

(b)
$$\frac{dy}{dt} - y^{5/3} = \frac{1}{1+t}, \quad y(0) = 1.$$

a) balance 1:
guess $|C| << |A| \sim |B|$

$\left|\frac{3}{t^3 + \cos t}\right| << |y'| \sim |y^2|$

for large t

setting $\frac{3}{} = 0$,

$t^3 + \cos t$

we have

$$y' = -y^2$$

$$\int -y^{-2} \frac{dy}{dt} dt = \int dt$$

$$\frac{1}{y} = y^{-1} = t + C$$

$$y = \frac{1}{t+C} \quad (t+c)^{-1}$$

$y(t)$ blows up when

$$t = -C$$

$$y(1) = 2$$

$$2 = \frac{1}{1+C}$$

$$2 + 2C = 1 \quad C = -\frac{1}{2}$$

$y(t) = \dfrac{1}{t - \frac{1}{2}}$ diverges @ $\underline{t = \frac{1}{2}}$

$t_0 = \frac{1}{2}$

however, the initial condition is at $t = 1$

so for large t, $y(t)$ decays to $0 \sim \frac{1}{t}$

$A = \dfrac{dy}{dt} = -y^2 \sim -\dfrac{1}{t^2}$

$B = y^2 \sim \dfrac{1}{t^2}$

$C = \dfrac{3}{t^3 + \cos t} \sim \dfrac{3}{t^3}$

$|C| << |A| \sim |B|$ ✓

correct for large t

balance 2: assume for large t

$$|A| \ll |C| \sim |B|$$

drop y'

$$y^2 = \frac{3}{t^3 + \cos t} \sim \frac{3}{t^3}$$

$$|y^2| \sim \frac{3}{t^3} = 3t^{-3}$$

$$|y| \sim \sqrt{3}\, t^{-\frac{3}{2}}$$

$$\frac{dy}{dt} \sim -\frac{3}{2}\sqrt{3}\, t^{-\frac{5}{2}} = -\frac{3\sqrt{3}}{2t^{\frac{5}{2}}}$$

$$\left|\frac{dy}{dt}\right| \sim \frac{3\sqrt{3}}{2t^{\frac{5}{2}}}$$

$$\left|\frac{3\sqrt{3}}{2t^{\frac{5}{2}}}\right| \cancel{\ll} \left|\frac{3}{t^3}\right| \sim \left|\frac{3}{t^3}\right|$$

$\uparrow$

larger at large t

invalid balance

balance 3:

$|B| \ll |A| \sim |C|$

$y^2 \qquad \frac{dy}{dt} \qquad \frac{3}{t^3 + \cos t}$

drop y^2

$$y' = \frac{3}{t^3 + \cos t} \sim \frac{3}{t^3}$$

at large t

$$\int \frac{dy}{dt}\,dt = \int 3t^{-3}\,dt$$

$$y(t) = -\frac{3}{2}t^{-2} + C$$

$y(1) = 2$

$$y(1) = 2 = -\frac{3}{2} + C$$

$$\frac{4}{2} + \frac{3}{2} = C = \frac{7}{2}$$

$$y(t) = -\frac{3}{2t^2} + \frac{7}{2} \sim \frac{7}{2}$$

$$|B| \sim \frac{49}{4}$$

$$|A| \sim 0$$

$$|C| \sim \frac{3}{t^3}$$

$$|B| >> |C|$$

invalid balance

$$\boxed{y(t) \sim \frac{1}{t} \text{ for large } t}$$

do matlab portion

b) $\overset{A}{\frac{dy}{dt}} - \overset{B}{y^{\frac{5}{3}}} = \overset{C}{\frac{1}{1+t}}$

$y(0) = 1$

balance 1:

guess

$|A| \ll |B| \sim |C|$

drop

$-y^{\frac{5}{3}} = \frac{1}{1+t} \sim \frac{1}{t}$

at large t

$-y \sim \frac{1}{t^{\frac{3}{5}}}$

$y \sim -\frac{1}{t^{\frac{3}{5}}} = -t^{-\frac{3}{5}}$

$$|A| \sim \frac{3}{5} t^{-\frac{8}{5}}$$

$$|B| \sim \frac{1}{t}$$

$$|C| \sim \frac{1}{t}$$

$$|A| << |B|$$

consistent balance

balance 2:

$$|B| << |A| \sim |C|$$

drop

$-y^{\frac{5}{3}}$

$$y' = \frac{1}{1+t}$$

$$\int \frac{dy}{dt} dt = \int \frac{1}{1+t} dt$$

$$y(t) = \ln(1+t) + C$$

$$y(1) = 0$$

$$0 = \ln(2) + C$$

$$C = -\ln(2)$$

$$y(t) = \ln(1+t) - \ln(2)$$

$$\sim \ln(t)$$

$$|B| \sim \ln(t)^{\frac{5}{3}}$$

$$|A| \sim \frac{1}{t}$$

$$|C| \sim \frac{1}{t}$$

$|B| \gg |A|$

invalid

balance 3:

$|C| \ll |A| \sim |B|$

drop $\frac{1}{1+t}$

$$y' = y^{\frac{5}{3}}$$

$$\int y^{-\frac{5}{3}} \frac{dy}{dt}\, dt = \int dt$$

$$-\frac{3}{2} y^{-\frac{2}{3}} = t + C$$

$$-\frac{3}{2}\frac{1}{y^{\frac{2}{3}}}=t+C$$

$$-\frac{3}{2}=y^{\frac{2}{3}}(t+C)$$

$$-\frac{3}{2(t+C)}=y^{\frac{2}{3}}$$

$$y(t)=-\frac{3^{\frac{3}{2}}}{2^{\frac{3}{2}}(t+C)^{\frac{3}{2}}}$$

$$=-\left(\frac{3}{2}\right)^{\frac{3}{2}}(t+C)^{-\frac{3}{2}}$$

$$y(1)=0$$

$$0=-\left(\frac{3}{2}\right)^{\frac{3}{2}}(C+1)^{-\frac{3}{2}}$$

$$=-\frac{\left(\frac{3}{2}\right)^{\frac{3}{2}}}{(C+t)^{\frac{3}{2}}}$$

$$=$$

C

$$\left|\frac{1}{1+t}\right| \sim \frac{1}{t}$$

A

$$\left|\frac{dy}{dt}\right| \sim \frac{3}{2}\left(\frac{3}{2}\right)^{3/2}$$

B

$$\left|y^{\frac{5}{3}}\right| \sim$$

- As needed, refine your approximate solution until it qualitatively agrees with the numerical solution.

(a)
$$\frac{dy}{dt} + y^2 = \frac{3}{t^3 + \cos t}, \quad y(1) = 2.$$

(b)
$$\frac{dy}{dt} - y^{5/3} = \frac{1}{1+t}, \quad y(0) = 1.$$

3. *Solving an initial value problem with Euler's method*
   In this problem you will write a program to solve the initial value problem
   $$\frac{dy}{dt} = 3t^2y^2, \quad y(0) = 0.3,$$
   in Matlab/Python by writing a function that implements Euler's method.

   (a) Create a new function called `eulermethod.m`. The inputs for this function are the initial condition $y(t_0) = y_0$, a final value of t which you can call t_f, and the step size h. With these inputs, `eulermethod.m` should do the following:

      i. Set y_0 as first element of the output array y.
      ii. Define an array t that goes from t_0 to t_f in increments of h.
      iii. Calculate subsequent values of the y array by implementing Euler's method.

   The outputs `eulermethod.m` should be two arrays: an array t as defined in part (a)ii above, and an array y which corresponds t script called `eulerdriver.m` that does the following:

      i. Calculates the solution to the initial value problem at $t = 1$ for step sizes $h = 0.001, 0.01, 0.1, 1.$
      ii. Creates a single figure with plots of the solution $y(t)$ versus t for each step size h.
      iii. Calculates the "exact" solution with `ode45`, and plots this solution on the figure from part (b)i.
      iv. For each value of h, calculates the error in Euler's method at $t = 1$, ie, the difference between the Euler solution at $t = 1$ and the solution obtained using `ode45` at $t = 1$.
      v. Creates a new figure with a plot of the error as a function of h. Is this plot consistent with your expectations? Check this by also plotting the error as a function of h on a log-log plot. Can you explain why a log-log plot may be helpful?

4. *Existence and uniqueness*
   For the initial value problems below, determine the longest interval in which the problem is certain to have a unique twice differentiable solution. Do not attempt to find the solution!

   (a) $ty'' + 3y = t, \quad y(1) = 1, \quad y'(1) = 2$
   (b) $(t-2)y'' + y' + (t-2)(\tan t)y = 0, \quad y(3) = 1, \quad y'(3) = 2$

5. *Linear algebra review*
   Consider the functions $f(t) = 3t - 5$ and $g(t) = 9t - 15$.

(a) Recall that a set of n vectors $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ in $\mathbb{R}^m$ is linearly independent if the only solution to $c_1\vec{v}_1 + c_2\vec{v}_2 + \ldots + c_n\vec{v}_n = \vec{0}$ is the trivial solution $c_1 = c_2 = \ldots = c_n = 0$.

Let $f(t)$ and $g(t)$ be vectors in the space of all continuous functions. Use this definition of linear independence to write down an equivalent system of linear equations $A\vec{c} = \vec{0}$ with $\vec{c} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$ and A to be determined. What is A? Then answer these questions:

i. What is $\det A$?

ii. Is A invertible?

iii. Do the columns of A form a linearly independent or linearly dependent set?

iv. Do the columns of A span $\mathbb{R}^2$?

(b) Is $f(t)$ a constant multiple of $g(t)$?

(c) Calculate the Wronskian of $f(t)$ and $g(t)$.

Explain how the calculations (a) - (c) above all verify that $\{f(t), g(t)\}$ is a linearly dependent set.

6. *Abel's Theorem*
Let y_1 and y_2 be solutions of the general linear homogeneous differential equation

$$L[y] = y'' + p(t)y' + q(t)y = 0,$$

where p and q are continuous on an open interval I. Then, as you'll show below, the Wronskian $W(y_1, y_2)$ is either always zero on I or never zero on I. This is called Abel's Theorem.

(a) Since y_1 and y_2 are two solutions to $L[y] = 0$, we have

$$\begin{aligned} y_1'' + p(t)y_1' + q(t)y_1 &= 0 \\ y_2'' + p(t)y_2' + q(t)y_2 &= 0 \end{aligned}$$

Multiply the first equation by $-y_2$ and the second equation by y_1, then add the equations together. In your resulting equation, group the terms with $p(t)$ together and the terms without $p(t)$ together.

(b) Recall that the Wronskian is $W(y_1, y_2)(t) = y_1 y_2' - y_1' y_2$. Differentiate this to get an expression for $W'(y_1, y_2)(t)$ in terms of y_1, y_1'', y_2, and y_2''.

(c) Now rewrite your equation in part (a) in terms of W and W'. You should have a first order linear (and separable) differential equation for W! Solve it! Your solution will have an arbitrary constant C. Notice that your solution for the Wronskian is either always zero (when $C = 0$) or never zero (when $C \neq 0$). This completes the proof.

7. If the functions y_1 and y_2 are linearly independent solutions of $y'' + p(t)y' + q(t)y = 0$, determine under what conditions the functions $y_3 = a_1y_1 + a_2y_2$ and $y_4 = b_1y_1 + b_2y_2$ also form a linearly independent set of solutions.

8. *Drill problems with the exponential ansatz*

(a) Find the general solution to the differential equation $y'' + 3y' + 2y = 0$.

(b) Find the general solution to the differential equation $y'' - 2y' - 2y = 0$.

(c) Find a differential equation whose general solution is $y(t) = c_1 e^{-t/2} + c_2 e^{-2t}$.

9. *Application: antibiotic production*
When you are sick your doctor may prescribe antibiotics to prevent or cure infections. The production of the antibiotic cycloheximide by *Streptomyces*[1] is typical of antibiotic production. During the production of cycloheximide, the mass of *Streptomyces* grows relatively quickly and produces little cycloheximide. After approximately 24 hours, the mass of *Streptomyces* remains constant and cycloheximide accumulates. However, once the level of cycloheximide reaches a certain level, extracellular cyclohexmide is degraded. One approach to alleviating this problem and to maximize cycloheximide production is to remove extracellular cycloheximide continuously.

The rate of growth of *Streptomyces* can be described by the logistic equation,

$$\frac{dX}{dt} = \mu_{\max}\left(1 - \frac{X}{X_{\max}}\right)X,$$

where X represents the mass concentration in g/L, $\mu_{\max}$ is the maximum specific growth rate, and $X_{\max}$ represents the maximum mass concentration[2].

(a) Find the solution to the initial value problem

$$\begin{aligned} \frac{dX}{dt} &= \mu_{\max}\left(1 - \frac{X}{X_{\max}}\right)X \\ X(0) &= 1 \end{aligned}$$

by first converting it to a linear equation with the change of variables $y = X^{-1}$. Please remember to apply the change of variables to both the differential equation and the initial condition.

(b) Experimental results have shown that $\mu_{\max} = 0.3\text{hr}^{-1}$ and $X_{\max} = 10$ g/L. Substitute these into the result obtained in part (a). In Matlab/Python, graph $X(t)$ on the interval $0 \le t \le 24$, and find the mass concentration at the end of 4, 8, 12, 16, 20, and 24 hours.

The rate of accumulation of cycloheximide is the difference between the rate of synthesis and the rate of degradation:

$$\frac{dP}{dt} = R_s - R_d.$$

It is known that $R_d = K_d P$, where $K_d \approx 5 \times 10^{-3}\text{hr}^{-1}$. Furthermore,

$$R_s = Q_{\text{po}} E X \left(1 + \frac{P}{K_I}\right)^{-1},$$

where Q_{po} represents the specific enzyme activity with $Q_{\text{po}} \approx 0.6$ g CH/g protein · hr and K_I represents the inhibition constant. E represents the intracellular concentration of an enzyme, which we assume to be constant.

[1] http://robdunnlab.com/projects/invisible-life/streptomyces/

[2] Dykstra and Wang, "Changes in the Protein Profile of *Streptomyces griseus* during a Cycloheximide Fermentation," *Biochemical Engineering V*, Annals of the New York Academy of Sciences, Vol. 56, New York Academy of Sciences (1987), pp. 511 - 522.

(c) For now, assume that P is finite. Use your solution from part (b) and show that, for large values of K_I and t,

$$\frac{dP}{dt} = 10Q_{\text{po}}E - K_dP.$$

Then solve the differential equation subject to the initial condition $P(24) = 0$.

(d) In Matlab/Python, graph $\frac{1}{E}P(t)$ on the interval $24 \leq t \leq 1000$.

(e) What happens to the net accumulation of the antibiotic as time increases? Is this consistent with finite P assumption in part (c)?

(f) If instead the antibiotic is removed from the solution so that no degradation occurs, what happens to the net accumulation of the antibiotic as time increases?